

IsPilot API Deployment Guide

This guide provides step-by-step instructions for deploying the IsPilot API to Google Cloud Run.

Prerequisites

Required Tools

- Docker (version 20.10+)
- gcloud CLI (version 500+)
- bash or PowerShell
- Git

Required Google Cloud Resources

- **Project:** corp-stro-salesinventory-prod
- **Service Account:** sa-ispilot-api@corp-stro-salesinventory-prod.iam.gserviceaccount.com
- **Artifact Registry:** ispilot-api in us-central1-docker.pkg.dev
- **Secret Manager:** ispilot-api-key secret
- **Firestore Database:** For user session storage
- **Vertex AI:** Reasoning Engine with ID 5375474415045705728

Required IAM Roles

The service account must have:

```
- roles/aiplatform.user
- roles/datastore.user
- roles/secretmanager.secretAccessor
- roles/logging.logWriter
- roles/monitoring.metricWriter
```

Setup: One-Time Configuration

Step 1: Create Service Account (if needed)

```
PROJECT_ID="corp-stro-salesinventory-prod"

gcloud iam service-accounts create sa-ispilot-api \
  --project="${PROJECT_ID}" \
  --display-name="IsPilot API Service Account"
```

Step 2: Grant IAM Roles

```
PROJECT_ID="corp-stro-salesinventory-prod"
SERVICE_ACCOUNT="sa-ispilot-api@${PROJECT_ID}.iam.gserviceaccount.com"

# Grant required roles
for role in \
  "roles/aiplatform.user" \
  "roles/datastore.user" \
  "roles/secretmanager.secretAccessor" \
  "roles/logging.logWriter" \
  "roles/monitoring.metricWriter"
do
  gcloud projects add-iam-policy-binding "${PROJECT_ID}" \
    --member="serviceAccount:${SERVICE_ACCOUNT}" \
    --role="${role}"
done
```

Step 3: Create API Key Secret

```
PROJECT_ID="corp-stro-salesinventory-prod"
SERVICE_ACCOUNT="sa-ispilot-api@${PROJECT_ID}.iam.gserviceaccount.com"

# Generate a secure random API key
API_KEY=$(openssl rand -hex 16)

# Create the secret
echo -n "${API_KEY}" | gcloud secrets create ispilot-api-key \
  --project="${PROJECT_ID}" \
  --replication-policy="automatic" \
  --data-file=-

# Grant the service account access to the secret
gcloud secrets add-iam-policy-binding ispilot-api-key \
  --project="${PROJECT_ID}" \
  --member="serviceAccount:${SERVICE_ACCOUNT}" \
  --role=roles/secretmanager.secretAccessor

echo "Created secret 'ispilot-api-key': ${API_KEY}"
```

Step 4: Create Artifact Registry Repository (if needed)

```
PROJECT_ID="corp-stro-salesinventory-prod"
REGION="us-central1"
REPOSITORY="ispilot-api"

gcloud artifacts repositories create "${REPOSITORY}" \
  --project="${PROJECT_ID}" \
```

```
--location="${REGION}" \  
--repository-format=docker
```

Deployment: Quick Start

Automated Deployment

```
# 1. Navigate to project root  
cd /path/to/ispilot-api  
  
# 2. Ensure script is executable  
chmod +x deploy.sh  
  
# 3. Run deployment script  
./deploy.sh
```

The script will:

1. Build Docker image
2. Configure Docker authentication
3. Push image to Artifact Registry
4. Deploy to Cloud Run
5. Verify deployment
6. Output service URL

Manual Deployment

If you need finer control:

```
PROJECT_ID="corp-stro-salesinventory-prod"  
REGION="us-central1"  
SERVICE_NAME="ispilot-api"  
REPOSITORY="ispilot-api"  
  
# 1. Build Docker image  
docker build -t us-central1-  
docker.pkg.dev/${PROJECT_ID}/${REPOSITORY}/${SERVICE_NAME}:latest .  
  
# 2. Configure Docker for Artifact Registry  
gcloud auth configure-docker us-central1-docker.pkg.dev --quiet  
  
# 3. Push image  
docker push us-central1-  
docker.pkg.dev/${PROJECT_ID}/${REPOSITORY}/${SERVICE_NAME}:latest  
  
# 4. Deploy to Cloud Run  
gcloud run deploy "${SERVICE_NAME}" \  

```

```

--project "${PROJECT_ID}" \
--region "${REGION}" \
--image "us-central1-
docker.pkg.dev/${PROJECT_ID}/${REPOSITORY}/${SERVICE_NAME}:latest" \
--platform managed \
--no-allow-unauthenticated \
--service-account "sa-ispilot-api@${PROJECT_ID}.iam.gserviceaccount.com" \
--memory 1Gi \
--cpu 2 \
--concurrency 100 \
--timeout 300 \
--set-env-vars \
  "GOOGLE_CLOUD_PROJECT=${PROJECT_ID}," \
  "GOOGLE_CLOUD_LOCATION=${REGION}," \
  "VERTEX_PROJECT_ID=${PROJECT_ID}," \
  "VERTEX_LOCATION=${REGION}," \
  "VERTEX_ENGINE_ID=5375474415045705728," \
  "FIRESTORE_COLLECTION=user_sessions," \
  "SESSION_TIMEOUT_HOURS=8" \
--update-secrets \
  "ISPILOT_API_KEY=ispilot-api-key:latest"

```

Verification

Get Service URL

```

PROJECT_ID="corp-stro-salesinventory-prod"
REGION="us-central1"
SERVICE_NAME="ispilot-api"

SERVICE_URL=$(gcloud run services describe "${SERVICE_NAME}" \
  --project "${PROJECT_ID}" \
  --region "${REGION}" \
  --format='value(status.url)')

echo "Service URL: ${SERVICE_URL}"

```

Test Health Endpoint

```

curl "${SERVICE_URL}/health"

# Expected response:
# {"status": "healthy", "timestamp": "2026-08-27T10:30:45.123456Z"}

```

Test Chat Endpoint

```
API_KEY=$(gcloud secrets versions access latest --secret=ispilot-api-key --
project=corp-stro-salesinventory-prod)

curl -X POST "${SERVICE_URL}/chat" \
  -H "Content-Type: application/json" \
  -H "X-API-Key: ${API_KEY}" \
  -d '{
    "user_id": "test-user",
    "message": "What is the inventory status?"
  }'
```

Check Service Status

```
PROJECT_ID="corp-stro-salesinventory-prod"
REGION="us-central1"
SERVICE_NAME="ispilot-api"

gcloud run services describe "${SERVICE_NAME}" \
  --project "${PROJECT_ID}" \
  --region "${REGION}"
```

View Logs

```
PROJECT_ID="corp-stro-salesinventory-prod"
REGION="us-central1"
SERVICE_NAME="ispilot-api"

# Recent logs
gcloud run services logs read "${SERVICE_NAME}" \
  --project "${PROJECT_ID}" \
  --region "${REGION}" \
  --limit 50

# Real-time logs
gcloud run services logs read "${SERVICE_NAME}" \
  --project "${PROJECT_ID}" \
  --region "${REGION}" \
  --limit 10 \
  --follow
```

CI/CD: Cloud Build Automation

The service uses Cloud Build for automated CI/CD. Each push to the main branch triggers a build pipeline.

Trigger Manual Build

```
gcloud builds submit \  
  --project=corp-stro-salesinventory-prod \  
  --config=cloudbuild.yaml \  
  --region=us-central1
```

View Build History

```
gcloud builds list \  
  --project=corp-stro-salesinventory-prod \  
  --limit=10
```

View Build Logs

```
BUILD_ID="your-build-id"  
gcloud builds log "${BUILD_ID}" \  
  --project=corp-stro-salesinventory-prod \  
  --stream
```

Rollback: Revert to Previous Deployment

Get Previous Revision

```
PROJECT_ID="corp-stro-salesinventory-prod"  
REGION="us-central1"  
SERVICE_NAME="ispilot-api"  
  
gcloud run revisions list \  
  --project="${PROJECT_ID}" \  
  --region="${REGION}" \  
  --service="${SERVICE_NAME}"
```

Deploy Previous Revision

```
PROJECT_ID="corp-stro-salesinventory-prod"  
REGION="us-central1"  
SERVICE_NAME="ispilot-api"  
REVISION_ID="ispilot-api-xxxxx"  
  
gcloud run services update-traffic "${SERVICE_NAME}" \  
  --project="${PROJECT_ID}" \  
  --region="${REGION}" \  
  --to-revisions "${REVISION_ID}=100"
```

Troubleshooting

"Docker push failed"

```
# Verify Docker authentication
gcloud auth configure-docker us-central1-docker.pkg.dev --quiet

# Check Docker daemon is running
docker ps
```

"Permission denied" on Cloud Run deployment

```
# Verify service account has required roles
gcloud projects get-iam-policy corp-stro-salesinventory-prod \
  --flatten="bindings[].members" \
  --filter="bindings.members:sa-ispilot-api"
```

"Secret not found"

```
# List available secrets
gcloud secrets list --project=corp-stro-salesinventory-prod

# Verify secret has the right accessor
gcloud secrets get-iam-policy ispilot-api-key \
  --project=corp-stro-salesinventory-prod
```

Health check fails immediately after deployment

1. Service needs time to start (30-60 seconds for first deployment)
2. Check service logs: `gcloud run services logs read ispilot-api --limit 50`
3. Verify all environment variables are set correctly
4. Verify Secret Manager secret is accessible

Cold start performance

- Multi-stage Docker build reduces image size
- Consider enabling minimum instances:

```
gcloud run services update ispilot-api \
  --project=corp-stro-salesinventory-prod \
  --region=us-central1 \
  --min-instances=1
```

Monitoring

View Metrics

```
PROJECT_ID="corp-stro-salesinventory-prod"
SERVICE_NAME="ispilot-api"

# View in Cloud Console
echo "https://console.cloud.google.com/run/detail/us-central1/${SERVICE_NAME}/metrics?project=${PROJECT_ID}"
```

Set Up Alerts

Use Cloud Monitoring to create alerts for:

- High error rates
- High latency (>1s)
- Service unavailability

Security

API Key Rotation

```
PROJECT_ID="corp-stro-salesinventory-prod"

# Generate new key
NEW_API_KEY=$(openssl rand -hex 16)

# Create new secret version
echo -n "${NEW_API_KEY}" | gcloud secrets versions add ispilot-api-key \
  --data-file=- \
  --project="${PROJECT_ID}"

# Cloud Run automatically uses the latest version (no redeployment needed)
echo "New API key: ${NEW_API_KEY}"
```

Service Account Key Management

Service account keys are NOT used for Cloud Run authentication. Instead, Cloud Run uses Workload Identity Federation for secure authentication.

Next Steps

1. Monitor service logs and metrics
2. Set up alerts for high error rates

3. Implement API rate limiting (if needed)
4. Set up automated backups for Firestore sessions
5. Configure custom domain (if needed)